Lecture 13

25.11.2025

We derived the weak/variational formulation of the 2D-problem

$$(*) \quad \begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \Gamma_1 \quad \text{(homogeneous Dirichlet boundary conditions)} \\ \frac{\partial u}{\partial \nu} &= 0 \text{ on } \Gamma_2 \quad \text{(homogeneous Neumann boundary conditions)} \end{aligned}$$

where $\Gamma_1 \cup \Gamma_2 = \Gamma = \partial\Omega$ with $\Gamma_1 \cap \Gamma_2 = \emptyset$.

Transformation was based on:

- trial space, for this concrete example $V_0 := \{w \in C^1(\bar{\Omega}) \mid w(x) = 0 \text{ on } \Gamma_1\}$
  ↳ here we look for our approximate solution
- test space, for this concrete example also V_0
  ↳ from here we take functions for multiplying ("testing") the PDE
- multivariate product rule: $\operatorname{div}(v \nabla u) = \nabla u \cdot \nabla v + v \underbrace{\operatorname{div}(\nabla u)}_{= \Delta u}$
- divergence theorem: $\underbrace{\int_\Omega \operatorname{div}(w)\, dx}_{\text{volume integral}} = \underbrace{\int_{\partial\Omega} \nu \cdot w \, ds}_{\text{surface integral}}$

  → the last two identities combined are also known as <u>Green's first identity</u>

$$\int_\Omega v \Delta w + \nabla v \cdot \nabla w \, dx = \int_{\partial\Omega} v (\nu \cdot \nabla w) \, ds$$

The result was the following weak form:

$$\boxed{\text{Find } u \in V_0 \text{ s.t. } \int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \quad \forall v \in V_0}$$

If there was an additional b.c. of the form $\frac{\partial u}{\partial \nu} + \sigma u = 0$ on Γ_3, we would have obtained the weak/variational formulation

$$\boxed{\text{Find } u \in V_0 \text{ s.t. } \int_\Omega \nabla u \cdot \nabla v \, dx + \int_{\Gamma_3} v \sigma u \, dA = \int_\Omega f v \, dx \quad \forall v \in V_0}$$

Weak derivatives and sobolev spaces

Up to now, for our trial and test spaces, we considered classically (continuously) differentiable functions $\rightsquigarrow$ too strong concept for finite elements since our typical candidates will look like this $\rightsquigarrow$ only "weakly differentiable"

Recall / note the space of square integrable functions over a domain $\Omega \subset \mathbb{R}^d$:

$$L^2(\Omega) := \left\{ v : \Omega \to \mathbb{R} \;\middle|\; \int_\Omega |v(x)|^2 \, dx < \infty \right\}$$

$\rightsquigarrow$ Hilbert space with scaler product and induced norm

$$\|u\|_{L^2} = \sqrt{\langle u, u \rangle_{L^2}} = \left(\int_\Omega |u(x)|^2 \, dx \right)^{1/2}$$

<u>Definition (weak derivative)</u> Let $u \in L^2(\Omega)$, where $\Omega \subset \mathbb{R}$ is an open interval. We say that $v \in L^2(\Omega)$ is <u>the weak derivative of u</u> if $\int_\Omega u(x) \varphi'(x) \, dx = -\int_\Omega v(x) \varphi(x) \, dx \;\; \forall \varphi \in C_c^\infty(\Omega)$.

<u>Note</u> if u is continuously differentiable and square integrable, then we obtain by integration by parts:

$$\int_\Omega u(x) \varphi'(x) \, dx = -\int_\Omega \underbrace{u'(x)}_{=v(x)} \varphi(x) \, dx + \underbrace{[u(x)\varphi(x)]_{\partial\Omega}}_{=0 \text{ since } v \in C_c^\infty(\Omega)!}$$

$\Rightarrow$ the classical derivative u' is the weak derivative!

But now we can also (weakly) differentiate other functions, e.g., $u : (-1,1) \to \mathbb{R}, x \mapsto u(x) = |x|$.

<u>Claim</u> $v(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$ is the weak derivative of u

$$\rightsquigarrow \int_{-1}^{1} u(x) \varphi'(x) \, dx = \int_{-1}^{1} |x| \varphi'(x) \, dx = \int_{-1}^{0} (-x) \varphi'(x) \, dx + \int_0^1 x \varphi'(x) \, dx$$

$$= \underbrace{[-x\varphi(x)]_{-1}^{0}}_{\substack{\text{vanishes since} \\ -x=0 \text{ and } \varphi(-1)=0}} - \int_{-1}^{0} (-1) \varphi(x) \, dx + \underbrace{[x\varphi(x)]_0^1}_{\substack{\text{vanishes since} \\ x=0 \text{ and } \varphi(1)=0}} - \int_0^1 (1) \cdot \varphi(x) \, dx$$

$$= -\int_{-1}^{0} v(x) \varphi(x) \, dx - \int_0^1 v(x) \varphi(x) \, dx = -\int_{-1}^{1} v(x) \varphi(x) \, dx \quad \forall \varphi \in C_c^\infty$$

Note: the value at $x = 0$ for v is not relevant

<u>Definition (weak derivative in $\mathbb{R}^d$)</u> Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with piecewise smooth boundary. Then $v \in L^2(\Omega)$ is the <u>weak derivative of u of order α</u> if

$$\int_\Omega v \varphi \, dx = (-1)^{|\varphi|} \int_\Omega u D^\alpha \varphi \, dx \quad \forall \varphi \in C_c^\infty(\Omega)$$

Definition (Sobolev space) The set

$$H^1(\Omega) := \{ u : \Omega \to \mathbb{R} \mid u \in L^2(\Omega),\ D^\alpha u \in L^2(\Omega) \text{ for } |\alpha| = 1 \}$$

is called (first-order) <u>Sobolev space</u>.

Let us return to our example from the beginning and its variational form

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \Gamma_1 \\ \frac{\partial u}{\partial \nu} &= 0 \text{ on } \Gamma_2 \end{aligned} \qquad \int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx$$

With our extended notion of differentiability, we can also use a slightly different trial/test space: $V_0 := \{ u \in H^1(\Omega) \mid u = 0 \text{ on } \Gamma_1 \}$

Using the notation $a(u,v) = \int_\Omega \nabla u \cdot \nabla v \, dx$ and $F(v) = \int_\Omega f v \, dx$ we can write the weak/variational form in a very compact way

$$\boxed{\text{Find } u \in V_0 \text{ s.t. } a(u,v) = F(v) \text{ for all } v \in V_0.}$$

Compare with the linear algebra problem:

$$\boxed{\text{Find } x \in \mathbb{R}^n \text{ s.t. } \underbrace{v^T A x}_{a(x,v)} = \underbrace{v^T b}_{F(v)} \text{ for all } v \in \mathbb{R}^n.}$$

Definition (Bilinear/linear form) Let V be a vector space over $\mathbb{R}$. Then $a : V \times V \to \mathbb{R}$ is called a <u>bilinear form</u> if

$$a(\alpha u + \beta v, w) = \alpha a(u,w) + \beta a(v,w),$$

$$a(w, \alpha u + \beta v) = \alpha a(w,u) + \beta a(w,v),$$

for all $u, v, w \in V$ and all $\alpha, \beta \in \mathbb{R}$. Similarly, $F : V \to \mathbb{R}$ is called a <u>linear form</u> if

$$f(\alpha u + \beta v) = \alpha f(u) + \beta f(v)$$

for all $u, v \in V$ and all $\alpha, \beta \in \mathbb{R}$.

Example (trial space $\neq$ test space) consider $-\Delta u = f$ in Ω, $u = g \neq 0$ on $\partial\Omega$.

Step 1: find any $u_g \in H^1(\Omega)$ s.t. $u_g(x) = g(x)$ on $\partial\Omega$

Step 2: find $u_0 \in V_0$ s.t. $a(u_0, v) = F(v) - a(u_g, v) =: F_g(v) \quad \forall\ v \in V_0$

vanishes on $\partial\Omega$

$\Rightarrow \quad u := u_0 + u_g \in H^1(\Omega)$, $u(x) = u_g(x) = g(x)$ on $\partial\Omega$

$\Rightarrow \quad a(u,v) = a(u_0 + u_g, v) = a(u_0, v) + a(u_g, v) = F_g(v) + a(u_g, v) = F(v) \ \forall v \in V_0$

Now: trial space is $H^1(\Omega)$, test space is V_0